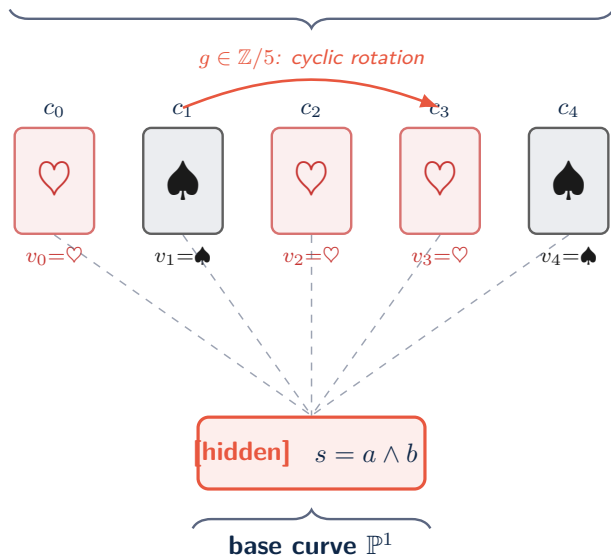


deck = covering curve C

Genus-0 example: 5 cards $\heartsuit \spadesuit \heartsuit \heartsuit \spadesuit$ on $C \cong \mathbb{P}^1$; $G = \mathbb{Z}/5$ acts by $x \mapsto \zeta x$, $\zeta^5 = 1$

Higher-genus realisations: S_5 on Bring's curve ($g = 4$), $S_5 \times S_5$ at $g = 173$



reconstruction

$\geq T$ shares give s
fewer than T :
 s stays uniform

$\pi : x \mapsto x^5$ on $\mathbb{P}^1$
(forget the rotation,
keep only $s = a \wedge b$;
algebraic relation,
not a data flow)

$s \in \{0, 1\}$: AND of two private bits, hidden until $\geq T$ cards combine